MATHS AA SL · Probability, simulation and decisions

Question booklet · 35 minutes · 28 marks

Answer all questions. A calculator is permitted. Show your reasoning. Give exact answers or round to three significant figures unless directed otherwise. Write on separate paper.

1. Two different faces [6]

A fair tetrahedral die has faces numbered 1 to 4. It is rolled twice. A technician records the larger score, using the common score if there is a tie.

(a) Find the probability that the score recorded is 3. **[3]**

(b) In 256 independent trials, give the expected number recording 3. A student predicts exactly this many. Comment. **[3]**

2. A fair simulation from digits [7]

A simulator produces independent random digits 0–9 with equal probabilities. A game needs three equally likely colours. Lin assigns 0–3 to red, 4–6 to blue and 7–9 to green.

(a) Calculate the probability of two reds in the first two colours under Lin's rule. Compare with a fair game. **[2]**

(b) Design a colour rule using the same generator that gives three equally likely colours. Justify it. **[3]**

(c) For your rule using one discarded digit, find the probability that at least one redraw is required while producing the first four colours. **[2]**

3. A second check on a parcel [8]

At a sorting depot, 12% of parcels need relabelling. A scanner flags 85% of those needing relabelling and 5% of all other parcels. A random parcel is selected.

(a) Find the probability it is flagged. **[2]**

(b) Given that it is flagged, find the probability it needs relabelling. Explain why the answer differs from 0.85. **[3]**

(c) The false-flag rate is adjustable. Find the largest false-flag rate for which at least 80% of flagged parcels need relabelling, with other rates fixed. **[3]**

4. Two gates at a venue [7]

At a concert, 120 people used the east gate and 80 the west gate during a recorded interval. Of these, 18 east-gate and 4 west-gate visitors waited more than five minutes.

(a) Estimate the probability of a long wait for a randomly selected visitor in that interval. **[2]**

(b) On another evening 40% use the east gate. Assuming the gate-specific rates stay the same, estimate the number of long waits among 600 visitors. **[3]**

(c) A manager attributes any change from 11% solely to faster service. Evaluate that conclusion. **[2]**

End of question booklet · 28 marks